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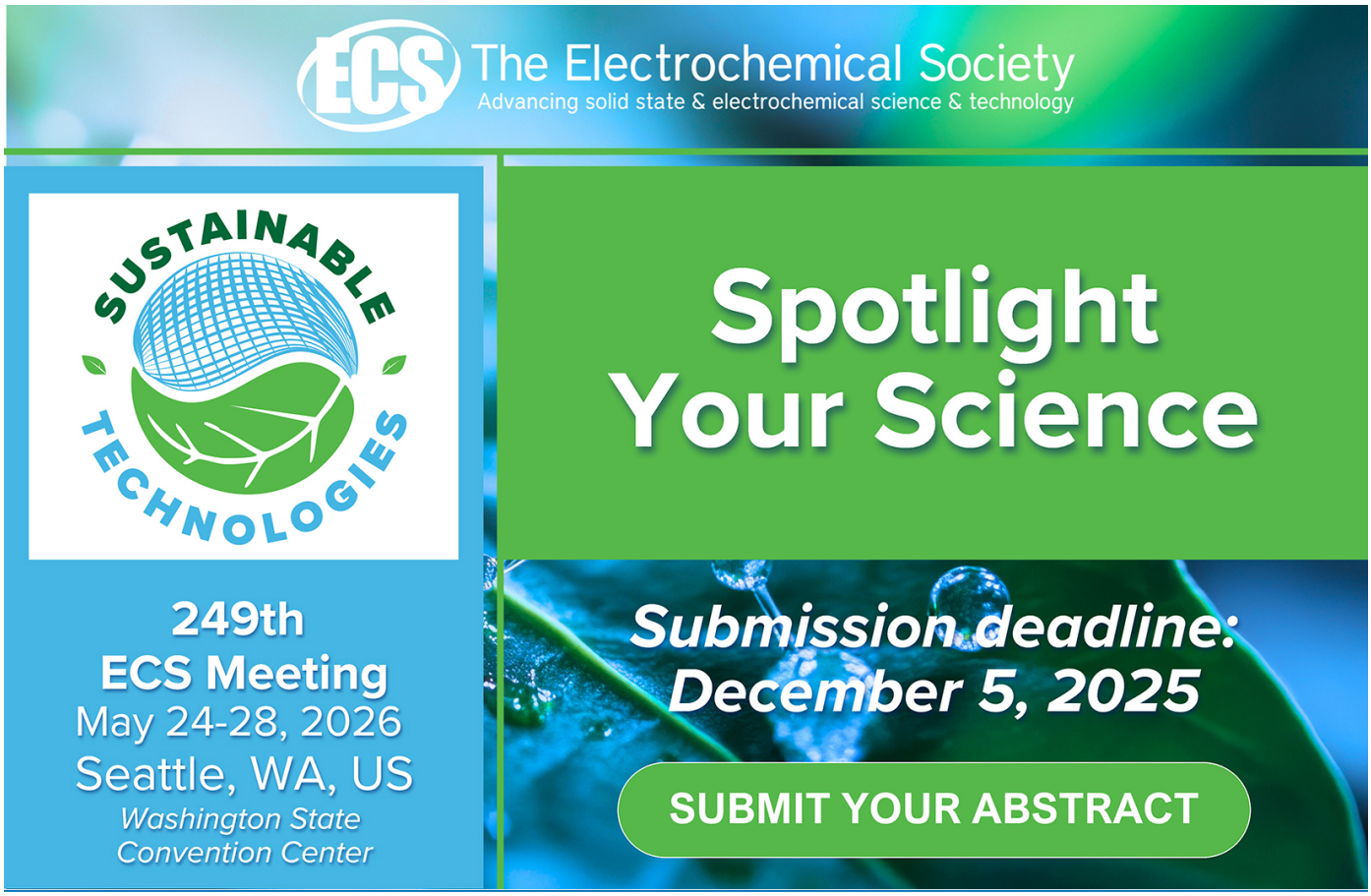

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LETTER

Future intensification of Northern Hemisphere monsoons due to declining remote aerosol pollution

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Supplementary material for this article is available online

Abstract

Anthropogenic aerosol emissions have significantly shaped historical monsoon precipitation, yet uncertainties persist in the projected response to future emissions. This study employs models contributing at least ten ensemble members to the Detection and Attribution Model Intercomparison Project—MIROC6 and CanESM5—to examine the mid-century response of the Northern Hemisphere (NH) summer monsoons to changes in aerosol burdens. We focus on a scenario characterized by an increase in aerosol burdens over South Asia, but strong reductions over the NH extra-tropics, since this is consistent with observed trends. These anomalous reductions induce an inter-hemispheric energy imbalance, prompting a large-scale response in the atmospheric meridional overturning circulation. The upper-tropospheric levels of the overturning circulation enhance heat transport towards the Southern Hemisphere, while the lower levels bring enhanced moisture convergence into the NH, leading to more rainfall across NH monsoon regions. Our findings highlight that global air pollution control measures may have wide-ranging impacts well beyond the aerosol source regions. For South Asia, these findings suggest that widespread remote aerosol reductions could offset the precipitation suppression from rising local aerosol pollution.

1. Introduction

Anthropogenic activities during the industrial era have substantially altered the planet's atmospheric composition impacting tropical monsoon systems (Wang *et al* 2021). While the influence of greenhouse gases (GHGs) on temperature and the hydrological cycle has been studied for the past fifty years (IPCC 1990), interest in the impact of anthropogenic aerosol emissions has surged in recent decades (Douville *et al* 2021).

Aerosols absorb and scatter shortwave radiation and alter cloud properties, which can reduce the meridional thermal contrast and weaken monsoon circulation (Charlson *et al* 1992, Penner *et al* 2001, Boucher *et al* 2013). Studies suggest that the Northern Hemisphere (NH) monsoon responds more sensitively to anthropogenic aerosols than to GHGs (Cao *et al* 2022). Rising anthropogenic aerosol emissions have been attributed as a key driver of the observed declines in the NH land monsoon precipitation (Polson *et al* 2014), and in the monsoon domains of West Africa, South, and East Asia since the 1950s (e.g. Rotstayn and Lohmann 2002, Ramanathan *et al* 2005, Biasutti and Giannini 2006, Lau and Kim 2006, Meehl *et al* 2008, Ackerley *et al* 2011, Bollasina *et al* 2011, Guo *et al* 2013, Krishnan *et al* 2016, Lin *et al* 2018, Undorf *et al* 2018, Bonfils *et al* 2020, Hirasawa *et al* 2020, Westervelt *et al* 2020).

Recent years have witnessed a growing interest in distinguishing the impacts of aerosols local to, and remote from, monsoon regions. Understanding this distinction is crucial because aerosol burdens and their associated radiative forcing exhibit

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heterogenous patterns across the world (Turnock *et al* 2020). Therefore, divergent country-scale mitigation policies on air pollution (Samset *et al* 2019, Persad *et al* 2022) may induce significant cross-border impacts on monsoon systems.

For example, while the historical drying and subsequent recovery of the West African monsoon have been largely attributed to remote aerosol emissions from North America and Europe (Rotstayn and Lohmann 2002, Ackerley *et al* 2011, Biasutti 2019, Giannini and Kaplan 2019, Monerie *et al* 2022), Shindell *et al* (2023) suggests that future changes in precipitation may witness a greater contribution from local aerosols. In contrast, studies of the South and East Asian monsoons have traditionally emphasized the importance of local aerosols (e.g. Ramanathan *et al* 2005, Lau and Kim 2006, Meehl *et al* 2008, Wang *et al* 2009, Ganguly *et al* 2012, Bollasina *et al* 2014, Salzmann *et al* 2014, Undorf *et al* 2018, Dong *et al* 2019, Wilcox *et al* 2020, Ayantika *et al* 2021, Singh *et al* 2023). Recent work, however, has highlighted significant contributions from remote aerosol sources even in these regions. For instance, Cowan and Cai (2011) demonstrated that remote aerosols can exacerbate the rainfall suppression induced by local aerosols for the Asian summer monsoon, while Ganguly *et al* (2012) reported nearly equal contributions from both local and remote sources in the South Asian Summer Monsoon (SAM). Undorf *et al* (2018) also found a detectable impact of remote aerosols on SAM precipitation. Furthermore, studies have revealed that remote *absorbing* aerosols can intensify SAM precipitation via atmospheric heating (Chakraborty *et al* 2014, Vinoj *et al* 2014, Krishnamohan *et al* 2021, Asutosh *et al* 2022), and others have reported significant impacts of remote *scattering* aerosols on the South and East Asian monsoons (Dong *et al* 2016, Shawki *et al* 2018). Guo *et al* (2016) and Asutosh *et al* (2022) found that remote sulfate and carbonaceous aerosols, respectively, play dominant roles in influencing SAM precipitation. However, these findings may have been influenced by limitations of utilizing atmosphere-only GCMs.

These works indicate that the future changes in West African monsoon, shaped mainly by remote aerosols in the past, may be increasingly driven by local emissions, while for South and East Asian monsoons, the understanding has shifted from the dominance of local aerosols to notable remote influence. This raises the critical question: How will the combined effects of local and remote aerosols influence tropical monsoon precipitation in the coming decades, especially over the SAM domain where their trends are anticipated to diverge (Wilcox *et al* 2020).

Here, we address this question by examining how regional monsoons respond to changes in the interhemispheric energy balance and associated shifts in the Intertropical Convergence Zone (ITCZ) (Ridley *et al* 2015). Previous research has shown that high NH aerosol concentrations post-1950s, caused an anomalous cooling in the hemisphere, driving a southward ITCZ shift (Schneider *et al* 2014) and weakening NH monsoons (NHMs) (Myhre *et al* 2014, Polson *et al* 2014). Future changes in aerosol forcing are therefore expected to affect the NHMs through similar pathways.

Projections indicate that scenarios with stringent aerosol emission controls exhibit enhanced monsoon precipitation over West Africa, South Asia, and East Asia (Acosta Navarro *et al* 2017, Wilcox *et al* 2020, Shindell *et al* 2023, Singh *et al* 2023). Yet, considerable spread persists in monsoon projections due to natural forcings and modes of variability (Hawkins and Sutton 2009, Ridley *et al* 2015, Huang *et al* 2020, Monerie *et al* 2022), and model limitations such as process representation and resolution (Stevens and Feingold 2009). Therefore, understanding the future monsoon response to changing aerosol emissions is a critical area of research.

This study adopts a holistic approach by analyzing both the overall NHM response and its monsoon subdomains. We focus on mid-century projections under the 'Middle of the road' CMIP6 scenario (Riahi *et al* 2017), where aerosol burdens over South Asia are projected to rise, while decreasing elsewhere in the NH.

An emerging body of work interprets global monsoons through an energetic lens, linking them to the interhemispheric energy gradient, and cross-equatorial energy transport. Building on the recently conceived 'energy flux' framework (e.g. Adam *et al* 2016, Boos and Korty 2016, Nicknish *et al* 2023, Chen *et al* 2024), we examine how changing aerosol burdens, particularly from remote sources, reshape tropical circulation and NHM precipitation. Nicknish *et al* (2023) demonstrated that remote forcings can drive future tropical rainfall changes via shifts in the energy flux equilibrium, while Chen *et al* (2024) highlighted the role of extratropical cloud feedbacks and Southern Ocean heat uptake in modulating the South Asian monsoon. Extending these insights, our study investigates how declining NH midlatitude aerosols generate interhemispheric energy imbalance, altering cross-equatorial energy and moisture transport, and intensifies NHM rainfall.

2. Models, data and methods

2.1. Data

We source data from the Detection and Attribution Model Intercomparison Project (DAMIP; Gillett *et al* 2016), which includes single-forcing experiments based on the shared socioeconomic pathways (SSPs) (Riahi *et al* 2017).

We use two DAMIP experiments: (1) HIST-AER, which includes only anthropogenic aerosols and precursors with other forcings fixed at pre-industrial levels, and (2) SSP-AER, which extends HIST-AER to 2100 following SSP2-4.5 aerosol trajectories.

2.2. Model and ensemble selection

To reduce uncertainties from internal variability, we select models with at least ten ensemble members, which can reduce the inter-ensemble spread by up to 75% compared to a single member (Monerie *et al* 2022). At the time of analysis, only three models met this criterion (see Text S2 and table S1): MIROC6 (Tatebe *et al* 2019), GISS-E2-G (Kelley *et al* 2020) and Can-ESM5 (Swart *et al* 2019). We excluded GISS-E2-G for reasons explained in section 3.1. Accordingly, we focus our analysis on the 20 ensemble members from MIROC6 and Can-ESM5.

2.3. Methods

Diagnostics were performed on the ensemble mean of MIROC6 and Can-ESM5 for the boreal summer season (June to September, JJAS). The 1984–2009 mean in HIST-AER defines the present-day climatology, the 2035–2059 mean in SSP-AER defines the mid-century climatology, and their difference represents the mid-century change for each ensemble member.

In the main text, we present results *averaged over all 20 members* (the 'grand-ensemble'), while model-specific ensemble means are presented in the Supplementary Information.

To interpret the NHM response in energetic terms, we analyze cross-equatorial atmospheric energy and moisture transport, along with Available Potential Energy (APE) (Text S3). For mechanistic interpretation, we analyze shifts in the mean meridional circulation using the zonally averaged meridional mass streamfunction (Oort and Yienger 1996, Dima and Wallace 2003, Hartman 2015).

Changes in summer monsoon precipitation (JJAS) are studied over the aggregate NHM domain (details in Text S1), and in the individual subdomains of West Africa (WAfM), South Asia (SAM), East Asia (EAM), and North America (NAmM) as defined by Wang *et al* (2014).

Furthermore, we assess the robustness of the projected change in precipitation in each monsoon domain by computing the signal-to-noise ratio, with 'signal' and 'noise' defined, respectively, as the future ensemble-mean change and standard deviation (SD) of the precipitation anomalies considering all 20 members. A signal-to-noise ratio greater than unity indicates that the forced change is greater than the combination of model uncertainty and internal variability (Meehl *et al* 2007, Monerie *et al* 2024).

3. Results

3.1. Changes in aerosol burdens and energy imbalance

Mid-century changes in AOD at 550 nm show a pronounced decrease over the NH mid-latitudes in both MIROC6 and Can-ESM5, particularly over North America, Europe, and East Asia, with a contrasting increase over South Asia (figure 1), which agree with recent trends (Samset *et al* 2019). Changes in aerosol burdens are marginal over the Southern Hemisphere (SH).

We ruled out GISS-E2-G since this model not only projects unrealistically high changes in AOD over Asia by mid-century (see figure S1 and Text S2B in the supplementary information), but also historical AOD trends (2002–2018) inconsistent with observations (Ramachandran *et al* 2022, Xiang *et al* 2023).

Changes in aerosol burdens can substantially alter the radiative energy input via changes in clear-sky albedo. As a result, the grand-ensemble-mean anomaly projects an increase in the net clear-sky shortwave flux across most of the NH, except South Asia, at both TOA and surface levels (figures 2(a) and (b)). This creates an inter-hemispheric asymmetry in radiation input that is further evidenced by the zonal mean anomalies shown in figures 2(c)–(e). These patterns are seen for both models, considering the respective 10-member ensemble means, but with higher magnitudes in CanESM5 (figures S2 and S3). A consequence of the energy imbalance is an anomalous warming across the NH midlatitudes, which sets up an interhemispheric temperature contrast at both the surface (figures S4(a), (b)) and in the troposphere (until 200 hPa; figure S5(a)). These results are consistent with Xiang *et al* (2023) and Samset *et al* (2025), who showed similar tropospheric warming signatures over the mid-latitudes arising from aerosol reductions (i.e. reductions in aerosol over East Asian regions).

3.2. Energy imbalance and atmospheric redistribution

We analyze the energy imbalance by studying anomalies in the energy flux potential (EFP; equation S1) in figure 3(a), where negative (positive) anomalies indicate atmospheric energy sources (sinks). Disaggregation by model reveals stronger sources and sinks in CanESM5 than MIROC6 (figure S6). Substantial negative EFP anomalies are seen over the NH midlatitudes, but little variation is seen over South Asia despite the increase in aerosol burden over this region. This is likely related to the *spatially widespread* reduction in aerosol burdens over the NH mid-latitudes spanning East Asia, Europe, and North

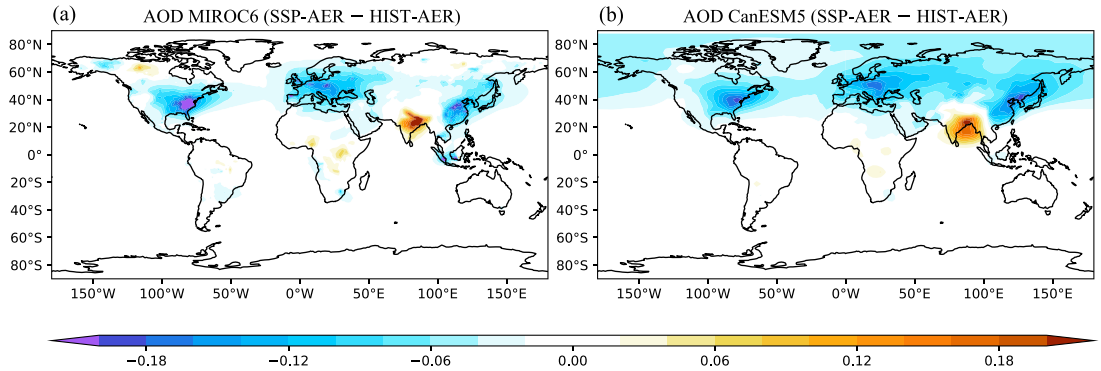

Figure 1. Boreal summer (JJAS) mid-century ensemble-mean anomalies in 550 nm AOD for (a) MIROC6, and (b) Can-ESM5. Anomalies representing future change are calculated for each member as the difference between the 2035–2059 climatology in SSP-AER and the 1984–2009 climatology in HIST-AER.

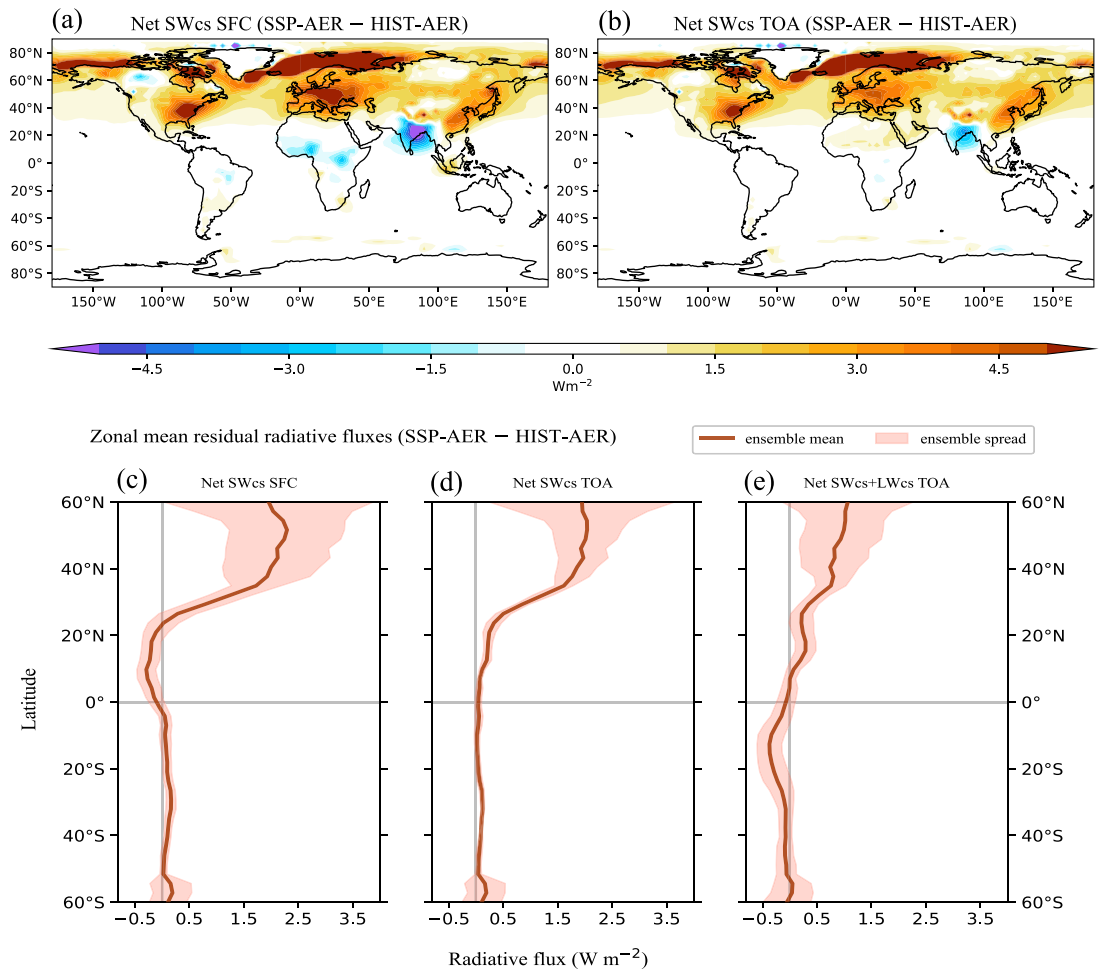

Figure 2. Boreal summer (JJAS) mid-century grand-ensemble-mean (i.e. mean over all 20 ensemble members) anomalies in (a) net clear-sky SW flux at the surface, and (b) net clear-sky SW flux at TOA. In (c)–(e), zonally averaged boreal summer (JJAS) grand-ensemble-mean anomalies in the net clear-sky (c) shortwave flux at the surface, (d) shortwave flux at TOA, and (e) net radiation (shortwave plus longwave) flux at TOA. Units are in W/m^2.

America, in contrast to the *spatially localized* increase in aerosol burdens over South Asia (figure 1). The differences in spatial extent of these changes manifest in the zonally averaged radiative response in the NH mid-latitudes (figures 2(c)–(e)) being several factors stronger in magnitude than the response in the 0–30° N bands corresponding to South Asia (figures 2(c)–(e)).

The anomalous sources in the NH and sinks in the SH elicit an atmospheric response in the form of an anomalous largescale *southward* cross-equatorial energy transport, with a grand-ensemble-median

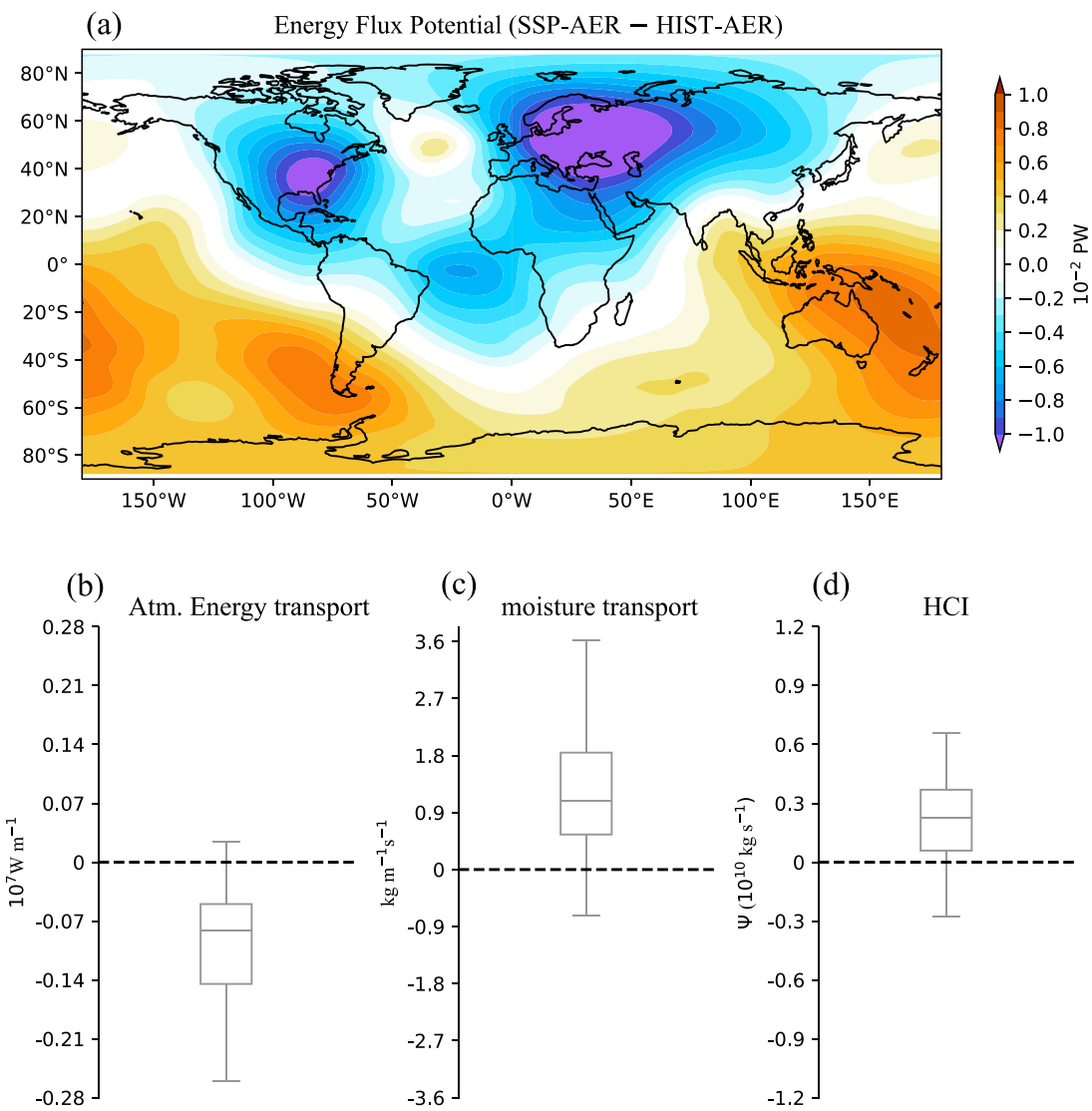

Figure 3. (a) Boreal summer (JJAS) mid-century grand-ensemble-mean anomaly in energy flux potential ($\times 10^{-2}$ PW). In (b)–(d): Box-whisker plots of the zonal mean boreal summer (JJAS) anomalies in (b) cross-equatorial heat transport ($\times 10^7$ W m^{-1}), (c) cross-equatorial moisture transport (kg m s^{-1}), and (d) Hadley circulation intensity (HCI, $\times 10^{10}$ kg s^{-1}), measuring by the overturning streamfunction. Boxes represent the interquartile range with the median indicated by the central horizontal line. Whiskers denote 1.5× interquartile range.

increase of about 800 000 W m^{-1} (figure 3(b)). Consistent with the stronger sources and sinks in CanESM5, the median increase in energy transport of about 1400 000 W m^{-1} is considerably larger than the median change of about 600 000 W m^{-1} in MIROC6 (figures S7(a) and (d)). Concurrently, there is an increase in the *northward* cross-equatorial moisture transport (grand-ensemble-median change of approximately 1 kg m^{-1} s^{-1}; same for both models (figures S7(b) and (e)), enhancing moisture supply to the NH (figure 3(c)).

To bolster the energetic argument, we analyzed changes in the APE (Lorenz 1955, Peixoto and Oort 1992) to show that an increase in APE strengthens the NHM circulation and precipitation (as explained in Text S3E). Thus, it reaffirms our inferences based on the EFP that the changes in NHM precipitation and circulation are forced by changes in inter-hemispheric energy imbalance.

3.3. Dynamical response: the Hadley circulation

The atmospheric adjustment to the energy imbalance occurs via a dynamical reorganization of large-scale circulations, notably the Hadley Cell, which links the changes in cross-equatorial energy and moisture transport. Whereas the enhanced southward energy transport occurs along its upper limb (Chen *et al* 2024), the enhanced northward moisture transport occurs along its lower limb.

We analyze changes in the Hadley circulation by studying changes in the zonally averaged meridional mass streamfunction (ψ), zonal wind (u), and meridional wind (v). For reference, the HIST-AER climatology exhibits the ascending and subsiding branches

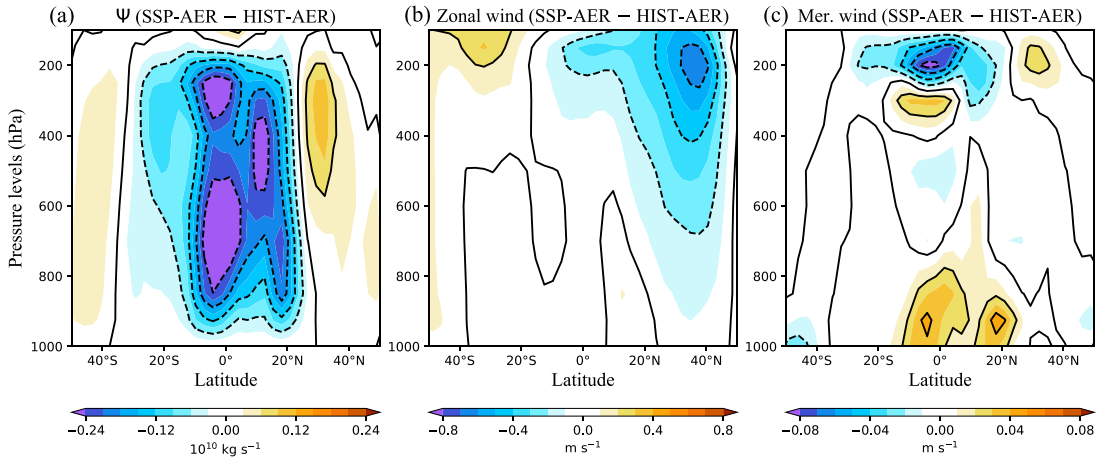

Figure 4. Zonally averaged boreal summer (JJAS) mid-century grand-ensemble-mean anomalies in (a) meridional mass streamfunction ($\times 10^{10}$ kg s^{-1}, see equation (S3)), (b) zonal wind (m s^{-1}), (c) meridional wind (m s^{-1}). In (a), positive values represent clockwise meridional circulation.

of the Hadley cell, along with a Tropical Easterly Jet (TEJ) at 200 hPa (figures S8(a)–(c) and S9(a)–(c)).

The increase in the intensity of Hadley circulation in the grand-ensemble-mean, defined as the maximum value of the meridional mass streamfunction at 500 hPa following Chemke and Yuval (2023), is about 0.25 $\times$ 10^{10} kg s^{-1} (figure 3(d)); the median change for the two models ranges between 0.2–0.3 $\times$ 10^{10} kg s^{-1} (figures S7(c) and (f)). A pronounced intensification is also seen in the ascending branch of the Hadley cell, in both the grand-ensemble-mean (figure 4(a)) and ensemble-mean for each model (figures S8(d) and S9(d)), with an associated strengthening of the upper-level northerlies and lower-level southerlies. The mid-century intensification is also evident in TEJ (figures 4(b), S8(e) and S9(e)).

The energy perspective thus reveals that the anomalous energy source in the NH mid-latitudes leads to a strong gradient in the EFP, strengthening the cross-equatorial heat transport in the upper troposphere via the increased NH meridional overturning circulation (figure 4). This is accompanied by enhanced low-level moisture transport and convergence into the NHM regions, thereby strengthening the boreal summer ITCZ.

3.4. Precipitation response

In line with the intensification of the ITCZ and the Hadley circulation, an increase in rainfall is projected over most NHM zones (figure 5(a)). Precipitation enhancement is most prominent for the WAfM with a rise over the entire domain. Over the SAM region, an increase is projected over the western coast and central landmass of India, and along the Himalayan foothills, while the increase in the EAM domain is largely confined to eastern China. These patterns are consistent in the ensemble means of the individual models (figure S10).

Domain-averaged JJAS precipitation anomalies for each ensemble member are shown in figure 5(b), revealing that most members project an increase in rainfall for the aggregate NHM and each subdomain. Notably, projections for the NHM domain exhibit lower spread than the subdomains with a grand-ensemble SD of about 0.09 mm d^{-1}, with the subdomains exhibiting a SD of at least 0.12 mm d^{-1} (table S2).

Among individual regions, WAfM and NAmM are distant from all major aerosol emission sources, yet projected to experience rainfall intensification by most members (figure 5(b)). The NAmM response is noteworthy since Douville *et al* (2021) assessed that rainfall would decline with warming in this region, although with low agreement among models. Our analysis reveals that an increase in NAmM rainfall is projected by most members in response to remote-aerosol-forcing. A possible outcome is that this aerosol-forced increase in precipitation may partially counteract the projected GHG-forced decrease over the NAmM domain. The SAM domain exhibits the smallest grand-ensemble-mean increase in rainfall of 0.11 mm d^{-1}, which is lower than the inter-member spread (grand-ensemble SD approx. 0.23 mm d^{-1}; table S2). It is evident from figure 5(b) that considerably more ensemble members disagree on the sign of change in SAM rainfall compared to other domains, which possibly arises from the uniquely competing effects of local and remote aerosol forcing for this region (figures 1(a) and (b)): the tendency for precipitation enhancement by declining remote aerosol burdens may be considerably counteracted by the tendency for rainfall suppression due to increasing local aerosol burden over this region (Ayantika *et al* 2021). However, quantitatively differentiating between the local and remote aerosol impacts on SAM circulation and precipitation is beyond the scope of this study, as it will require carefully

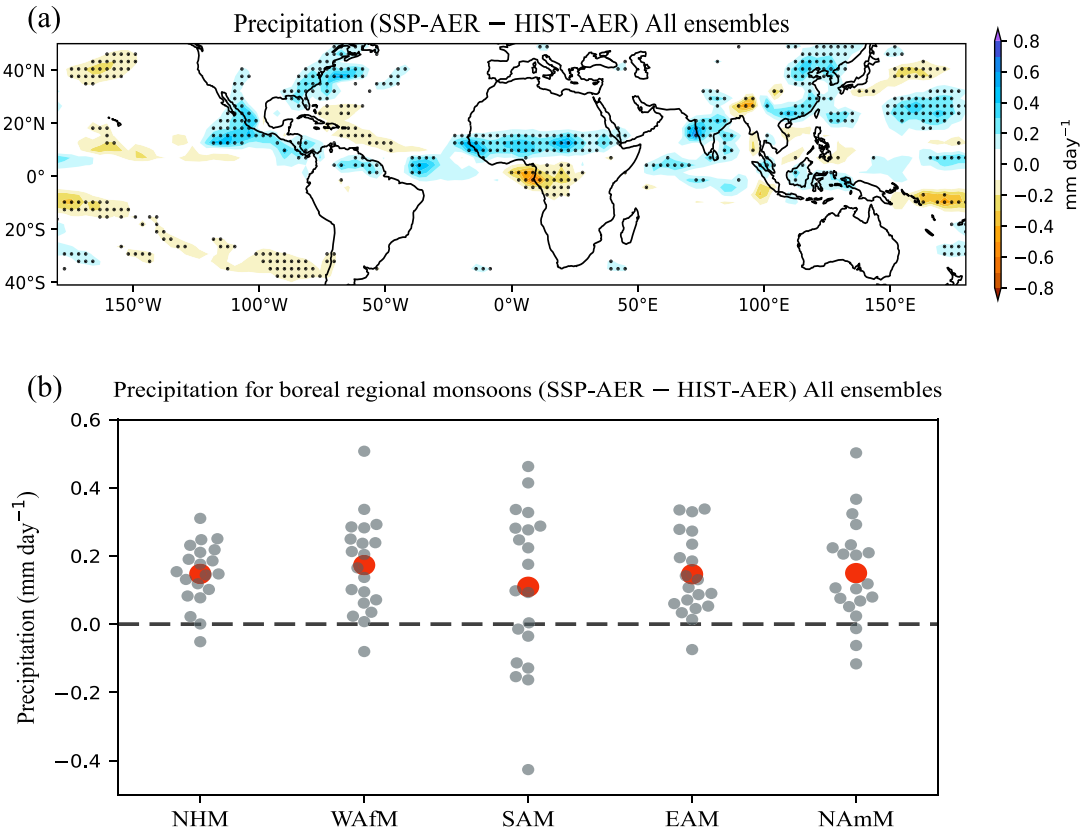

Figure 5. (a) Boreal summer (JJAS) mid-century grand-ensemble-mean precipitation anomaly (in mm/day), with stippling indicating significance at the 90% confidence level according to a two-tailed student's t-test with unequal variances. (b) Domain-averaged precipitation anomalies (in mm/day) for all 20 ensemble members for the North Hemisphere (NHM), West African (WAfM), South Asian (SAM), East Asian (EAM), and North American (NAmM) monsoon domains. Grand-ensemble-mean anomalies are indicated by red dots.

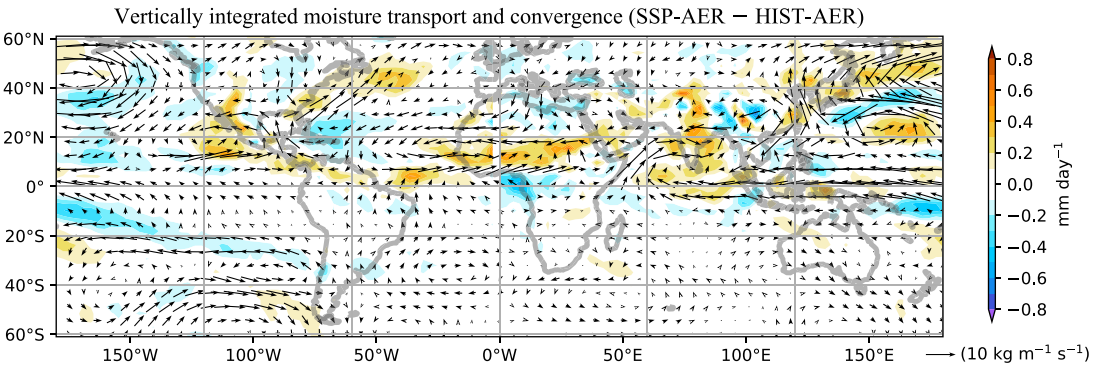

Figure 6. Boreal summer (JJAS) grand-ensemble-mean anomaly of vertically integrated moisture transport (vectors, $\times 10$ kg m^{-1} s^{-1}) and vertically integrated moisture convergence (VIMC; shading, $\times 10^{-6}$ kg m^{-2} s^{-1}). Vertically integrated moisture convergence is positive for net moisture convergence into the atmospheric column.

designed experiments such as those in the Regional Aerosol Model Intercomparison Project (RAMIP, Wilcox *et al* 2023).

To understand these regional hydrological responses, we examine the vertically integrated moisture transport and convergence in figure 6. Anomalous cyclonic westerlies over West Africa, intruding deep to occupy the entire tropical African landmass, lead to enhanced moisture transport from the adjacent oceanic regions, causing strong moisture convergence (shading). Over South Asia, strong southwesterly anomalies over the Arabian Sea, in conjunction with southeasterly anomalies over the central Indian plains, cause large-scale moisture convergence over the Indian region. These patterns are consistent for both models (figure S11).

The increased regional monsoon precipitation over the NH further influences subtropical circulations, evidenced by a pronounced strengthening of the North Pacific High and the Azores High (Rodwell and Hoskins 2001). Enhanced easterlies over the equatorial Pacific may reflect a Kelvin-wave response (Gill 1980, Rodwell and Hoskins 2001, Terray *et al* 2018) to the anomalous diabatic heating over the South and East Asian monsoon regions.

In summary, the anomalous rainfall and circulation responses indicate a strong sensitivity of NHMs to reduced aerosol emissions over the NH extratropics. These results corroborate previous modeling studies (e.g. Dong *et al* 2016, Guo *et al* 2016, Acosta Navarro *et al* 2017, Shawki *et al* 2018, Krishnamohan *et al* 2021, Asutosh *et al* 2022, Cao *et al* 2022), while expanding the focus to include both the aggregate NHM and regional monsoon systems.

4. Summary and discussion

This study utilizes DAMIP simulations to examine the mid-century response of NH summer monsoons to changes in aerosol emissions under the SSP2-4.5 scenario in which anthropogenic aerosol burdens decrease over the US, Europe, and East Asia but rise over South Asia.

We find a robust (discussed below) aerosol-induced increase in precipitation over the West African and North American monsoon regions, despite their distance from major aerosol sources. In South Asia, we found that the anticipated precipitation suppression from increasing local aerosol emissions is more-than-compensated by the precipitation enhancing effects of declining remote aerosols. This leads us to conclude that remote reductions in aerosol burdens will profoundly impact tropical rainfall in the coming decades.

Energetically, the reduced clear-sky albedo from the *spatially widespread* reductions in NH midlatitudes result in the NH behaving as a strong energy source region dominantly forcing dynamical changes. We find that the increase in aerosol burdens over South Asia are too *spatially localized* to counteract the strong midlatitude forcing or significantly affect the broader Hadley Cell response. The consequent anomalous energy absorption in the NH induces southward energy transport via the upper limb of a strengthened Hadley cell. This concurrently increases northward moisture transport along the Hadley Cell's lower limb into the NHM regions. Thereby, the boreal summer ITCZ is strengthened over the NH, and precipitation intensifies over the entire domain. These findings are broadly in line with earlier studies studying albedo changes resulting from other factors like clouds (e.g. Voigt *et al* 2014), ice (Chiang and Bitz 2005), stratospheric aerosols (Yoshimori and Broccoli 2008, Haywood *et al* 2016), and land use and land cover (Laguë *et al* 2021) that can modify the radiation budget, causing changes in circulation and the ITCZ's position.

It is important to recognize that projections are subject to multiple uncertainties arising from internal climate variability and differences in climate sensitivity (Hawkins and Sutton 2009, Huang *et al* 2020). While our analysis is confined to MIROC6 and CanESM5 for reasons explained in sections 2.2 and 3.1., these two models differ considerably in their aerosol and cloud treatments, as well as their effective climate sensitivities. Specifically, CanESM5 employs a prognostic scheme for bulk concentrations of natural and anthropogenic aerosols and another for cloud microphysics, yielding a high value for the effective climate sensitivity of 5.65 K (Virgin *et al* 2021). MIROC6 utilizes the Spectral Radiation Transport Model for Aerosol Species to simulate detailed aerosol processes and distinguishes between different aerosol species, while also incorporating a prognostic cloud scheme augmented by an explicit shallow convection parameterization, resulting in a lower effective climate sensitivity of about 2.8 K (Tatebe *et al* 2019). Despite these differences between models, projected changes in Hadley cell intensity, and heat and moisture transports, are consistent across most ensemble members of both models for all monsoon regions. Furthermore, we find that the ensemble-mean precipitation increase for the aggregate NHM domain of approx. 0.15 mm d^{-1} exceeds the inter-member SD of about 0.09 mm d^{-1}, which is smaller than the spread for subdomains. This suggests an emergence of the forced signal above the model uncertainty and internal variability at the hemispheric scale (see section 2.3). Furthermore, the mean changes in the West African and East Asian domains are similarly larger than the corresponding SD, whereas the change over the North American domain equals the inter-ensemble SD. The only exception is the South Asian domain, where the mean change (0.11 mm d^{-1}) is smaller than the inter-ensemble SD (0.23 mm d^{-1}), possibly on account of the uniquely counteracting trends in local and remote aerosol forcing.

It is important to note, however, that while using the inter-ensemble SD from forced simulations as a measure of internal variability is a useful diagnostic, it does not capture the full spectrum of natural variability present in a pre-industrial control simulation. Despite this limitation, the fact that the forced signal exceeds the SD in most monsoon regions supports the conclusion that the projected precipitation changes are expected to emerge from the internal variability of the climate system (Huang *et al* 2020, Singh *et al* 2023). A more comprehensive quantification of the roles played by model climate sensitivity, resolution, and process representation (e.g. aerosol–cloud interactions) will ultimately require carefully

designed experiments using a broader array of models (Wang *et al* 2021). Expanded participation in intercomparison projects such as DAMIP will be central to this effort, to better understand the influence of aerosol processes on monsoon dynamics. Another way forward is to make use of recently available Large Ensemble Single Forcing Model Intercomparison Project to delineate the effects of the individual external forcings (Smith *et al* 2022).

Additionally, it may be noted that our analysis is focused on only the SSP2-4.5 scenario for conceptual and pragmatic reasons. While, in principle, it would be viable to consider SSP1 or SSP5 for analysis, in practice, the strong air-quality assumptions inherent to SSP1 yield aerosol trends that are inconsistent with recent observations (Samset *et al* 2019), making it unsuitable for our analysis. On the other hand, the emissions trajectories and mid-century aerosol burdens under SSP5 closely mirror SSP2 trajectories and burdens (Wilcox *et al* 2020), signifying that results from analyzing SSP5 would differ little from the analysis presented above. Besides these conceptual reasons to favor SSP2 for analysis, we are also constrained by the DAMIP project only specifying and archiving SSP2-4.5 projections (Gillett *et al* 2016).

In summary, our results reinforce the assertion that aerosols constitute a major climate risk factor, especially for West Africa and South Asia. The fact that aerosol-induced perturbations can significantly modulate monsoon precipitation on a hemispheric scale underscores the importance of integrating aerosols into broader climate risk assessments (Persad *et al* 2022). Ultimately, reducing these risks will necessitate enhanced transboundary cooperation in developing effective air pollution mitigation policies. Such concerted efforts are essential for managing the regional and global impacts of aerosol pollution on climate.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://aims2.llnl.gov/search.

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